



Crater identification simulation using LiDAR on Lunar rover

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ARTICLE INFO

Keywords:

LiDAR
Environment Perception
Systematic error
Crater identification
Terrain frequency

ABSTRACT

Crater identification is an important task of environment perception, which is benefit for path planning and safe campaign of Lunar rover. In order to overcome the partial missing of point cloud obtained by LiDAR, a three-dimensional data-based crater identification method is proposed to complete the crater by filling in the blind zone for high probability detection, which can be evaluated by completion rate and detection probability of the negative obstacle area in the occupancy grid map, respectively. To obtain a reasonable identification of the craters under various terrain conditions, three parameters related to the terrain and the sensor are varied for evaluating the consequence of crater identification: the terrain frequency, the ranging errors and the angle errors. In simulation experiments, craters with various parameters locate on rolling terrain. The parameters settings used in simulation experiments include the diameter of craters (2 m to 20 m), terrain frequency ($>20 \text{ m}^{-1}$), ranging error ($<0.07 \text{ m}$) and vertical angle error ($<0.3^\circ$). The simulation experimental results showed that the proposed method achieves the completion rate of exceeding 84.18 % for providing the complete craters, and the detection probability more than 80.58 % to create the safe occupancy grid map for path planning. This three-dimensional environment perception simulation for crater identification will be instructive for in-situ surface data evaluation and Lunar exploration.

1. Introduction

Lunar rover works in the Lunar environment for a long time and a large range. The autonomous navigation of Lunar rover is the basis for achieving its work, which requires a lot of testing of its performance. Due to the high risk and high value of Lunar exploration missions, large-scale field test of the Lunar environment costs too much. Thus, establishing a three-dimensional environment perception analysis offers a bond between the sensor parameters and the complex Lunar rugged environment, which helps us to understand if the perception algorithm implemented on Lunar rover is suitable [1,2].

Light detection and ranging (LiDAR) technology is a reliable and efficient approach for Lunar rover to perceive the surroundings [3–5]. In the pre-21st century researches, the positioning system employed stereo-imaging, encoders, inclinometers, a compass, and a turn-rate sensor to maintain the position and orientation of the rover as it traversed [6,7]. The NASA Spirit and Opportunity rovers, as well as China's Yutu, have all used stereo-imaging to create three-dimensional images over short distances for obstacle detection [8–10]. However, compared to LiDAR, stereo-imaging has a shorter detection range and more easily

affected by light conditions. In contrast, among various types of LiDAR, the solid-state LiDAR uses solid state components without moving parts [11], so the advantages of high vibration resistance and compact volume allows itself to fulfil the exploration in the Lunar rugged environment. In the next decades of various missions, LiDAR will be installed as an important sensor in the Lunar rover. An important motivation of using LiDAR is to develop a guidance system for a surface rover that could explore such permanently shadowed locations without optical help on the polar terrains of the Moon [12]. The points density and the points accuracy are heavily determined by the LiDAR system parameter, which further impact on the environment perception. There is still a need for more detailed scientific investigation into LiDAR system design parameters and operations that impact environment perception for Lunar rovers. Consequently, this study focuses on the environment perception performance of Lunar rovers changed with the quality of point cloud and three-dimensional environment map.

The three-dimensional (3D) environment perception of Lunar rover mainly includes the environment simulation and sensor system simulation. Researchers have carried out a series of studies on these. The simulation of Lunar surface 3D environment is the base of Lunar rover

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<https://doi.org/10.1016/j.measurement.2023.112550>

Received 11 October 2022; Received in revised form 27 December 2022; Accepted 27 January 2023

Available online 31 January 2023

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simulation system. In 2019, Allan et al. applied ray-casting method to construct the Lunar surface environment based on the Digital Evaluation Model (DEM) with resolution of 3–5 cm [13]. In 2019, Win et al. constructed the virtual Lunar environment with GNU image manipulation program using available image data of the Mare Tranquillitatis crater [14]. In 2021, Yu et al. built a lunar rover training system by using Gazebo 3D simulation environment, which loads the scaling model of the real terrain data of the moon by the Chinese Chang'e-2 photography with resolution of 20 m [15]. In terms of sensors, in 2015, Bualat et al. used binocular camera to construct visual odometer to track the movement of Astrobee, and used depth camera to construct map for path planning [16]. As the LiDAR applied in the lunar rover, in 2021, Gargiulo et al. estimated trajectory of the rover through the LiDAR data processing [17]. However, one of the main deficits in the previous works for the environment perception of the Lunar rover is the lack of reconfigurable three-dimensional data for Lunar terrain. The DTM based crater analysis at shadowed polar regions have been done using topographic data from orbit, but its data scale is not suitable for validating the path planning method to navigate around the small craters. In our work, based on the basic crater model in PANGU (Planet and Asteroid Natural Scene Generation Utility) [18,19], a parabolic curve was adopted to approximately construct the crater models with the diameter ranging from 2 m to 25 m, forming the reconfigurable Lunar terrain.

The biggest difficulty in 3D environment perception for Lunar rover is finding the negative obstacles by special sampling the complicated rugged environment for safe path planning. The lunar rover is affected by many environmental factors on the lunar surface during its mission. This becomes more challenging in the lunar environments due to the higher probability of finding negative obstacles compared with structural earth environments as there are a great many craters on the Lunar surface. Negative obstacles are more difficult to detect than positive obstacles due to limited visibility which further decreases as the distance from the negative obstacles to the vehicle increases [20,21]. In 2016, Shang et al presented a feature fusion-based algorithm for negative obstacle detection with LiDAR sensors and proved the robustness and stability [22]. In 2020, Zhong et al presented an adaptive negative obstacle detection method for autonomous vehicles using multiple side-mounted LiDARs [23]. In 2021, Daniel et al developed an analytical model for predicting the ability of UGV mounted LiDAR sensors to detect negative obstacles, showing that detection ranges for UGV are found to be less than 10 m [24]. However, they did not estimate the shape or the center of the negative obstacles, which are challenging for the Lunar rover. On these grounds, a crater detection and completion method is proposed to improve the identification rate and further reduce the risk ability of the Lunar rover path planning.

Two main issues concern the existing studies: (1) the influencing factors involved in three-dimensional crater identification is not figured out or analyzed, which greatly effects on perception environment performance; (2) the accuracy and completeness of crater identification need to be improved. Aiming to solve these problems, this paper reports a study on a LiDAR perception model involving system parameters and errors, employing the indicators of crater identification to embody the influence of different three-dimensional data quality. The paper loads the terrain data of the Lunar surface, and creates the simulation prototype of the Lunar rover and LiDAR in this environment to verify and evaluate different terrain complexity and LiDAR influencing factors. This research aims to estimate the influence of characteristics of LiDAR system and terrain on lunar craters identification.

The paper is organized as follows: Section 2 describes a LiDAR perception model and generation method for point cloud of Lunar, algorithm of obstacles identification and its evaluation indicators. Results demonstrate the influence of the system design parameters in a LiDAR perception model on point cloud in Section 3. A series of simulation are performed to verify crater identification influenced by terrain frequency and point cloud quality. The significance of the critical parameters on the LiDAR system design is discussed in Section 4. Conclusions are

presented in Section 5.

2. Methodology

One goal of LiDAR-based Lunar rover is to perceive the terrain of Lunar with traversable area and obstacles for further path planning. The challenge is to build and optimize rough Lunar terrain map based on discrete and incomplete point cloud. This section describes the common LiDAR model, which can be suitable for most kinds of LiDAR, such as flash LiDAR and MEMS (Micro-Electro-Mechanical System) LiDAR. Lunar terrain model and how the proposed methods collect point cloud efficiently, detect and complete the obstacles and finally output the occupancy grid map. The overall methodology is shown in Fig. 1.

2.1. LiDAR perception model

To analyze the point cloud quality, it's necessary to build the LiDAR perception model containing the system parameters and errors. LiDAR emits high power in short pulses, causing an increased return signal to a receiver, while the ranging information for every pixel in the two-dimensional field of view. The LiDAR coordinate system $O_L - X_L Y_L Z_L$ is established with the geometric center of the LiDAR as the origin (O_L). The transmitted and received signals can be equaled as that LiDAR emits lasers at the same time in both vertical and horizontal directions. The unit vector of column i ($i = 1, 2, \dots, M_h$) and row j ($j = 1, 2, \dots, N_v$) laser in the LiDAR coordinate system can be determined by the vertical field of view angle (α_v , unit: rad), the number of vertical pixels (N_v), the horizontal field of view angle (β_h , unit: rad) and the number of horizontal pixels (M_h).

The horizontal angle θ_i is the angle between the projection of laser ray direction on plane $X_L O_L Y_L$ and X-axis, which is determined by β_h and M_h , and can be calculated as

$$\theta_i = -\frac{\beta_v}{2} + (i-1) \cdot \frac{\beta_v}{M_h} \quad (1)$$

The vertical angle φ_j is the angle between the laser emission direction and the plane $X_L O_L Y_L$, which is determined by α_v and N_v , and can be calculated as

$$\varphi_j = -\frac{\alpha_v}{2} + (j-1) \cdot \frac{\alpha_v}{N_v} \quad (2)$$

Laser vector $\vec{L}_{ij}^L$ is the unit vector centered at the origin of the LiDAR coordinate system. Then the laser vector $\vec{L}_{ij}^L$ can be calculated as

$$\vec{L}_{ij}^L = \begin{bmatrix} \cos\theta_i \cos\varphi_j \\ \sin\theta_i \cos\varphi_j \\ \sin\varphi_j \end{bmatrix} \quad (3)$$

So once get the laser unit vector of each direction, the LiDAR scanning points can be solved as the intersection P_{ij} of the laser with the

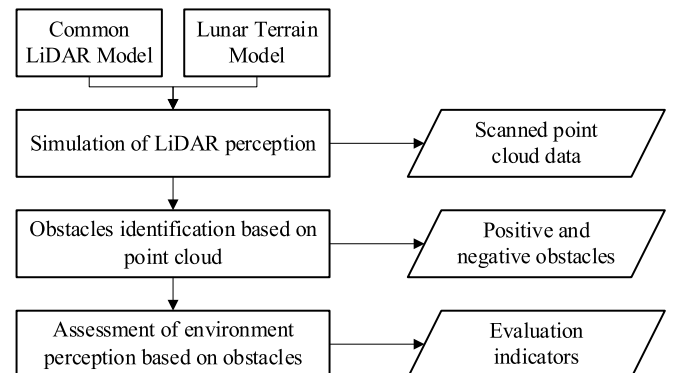


Fig. 1. Flow chart of the research methodology.

environment surface and distance R_{ij} , which can be referred to our previous work [25].

Based on the LiDAR perception model, the quality of the point cloud changes with the variation of its parameters. The quality of point cloud is mainly divided into two aspects, namely density and accuracy. Density is the number of points per unit area on the target terrain, reflecting the sparsity and richness of point cloud. The point cloud density decided by the laser spatial distribution, which is affected by the vertical field of view (FOV) angle (α_v), the number of vertical pixels (N_v), the horizontal field of view angle (β_h) and the number of horizontal pixels (M_h) of LiDAR system. The density of the point cloud becomes higher with more pixels and smaller FOV.

The point cloud accuracy e^2 represents the spatial error between the LiDAR measuring point and the solution point:

$$e^2 = \Delta R_{ij}^2 + \Delta \theta_i^2 \cdot L^2 \cos^2 \varphi_j + \Delta \varphi_j^2 \cdot R_{ij}^2 \quad (4)$$

The accuracy of point cloud is mainly affected by ranging error ΔR_{ij} and pointing error, which includes horizontal angle error $\Delta \theta_i$ and vertical angle error $\Delta \varphi_j$. The above errors may be caused by the internal mechanism of the LiDAR, system installation and the movement of the platform. The source of the errors is not discussed in this paper, and only the errors of Gaussian white noise are expressed as $\Delta \theta_i$, $\Delta \varphi_j$ and ΔR_{ij} .

2.2. Lunar terrain model

In addition to the point cloud quality caused by system parameters and errors, the topographic fluctuation of the Lunar terrain influences the results of environment perception. Therefore, it is significant to discuss the relationship between terrain and sensor spatial sampling by terrain simulation method. For the existing DEM of the Moon, the resolution is about 70 m, so it is difficult to obtain the detailed features of obstacles. To obtain a more realistic Lunar point cloud, the Lunar geometric topography needs to be analyzed and simulated and to obtain the reconfigurable three-dimensional terrain model.

The basic obstacles on the Lunar surface are craters and rocks. Lunar craters are circular depressional formations of different sizes on the Lunar surface, formed by the meteoroid impacting on the Lunar surface. Lunar craters are mainly composed of crater rim and crater depression, and the simplified crater model is shown in Fig. 2.

The simplified Lunar crater model is constructed by the surface of two parabolas rotating around the z axis. The crater rim model is calculated by formula, and the main body of the crater is calculated by formula.

$$z_{crt}(x, y) = h_{crt} - \frac{h_{crt}}{r_{crt}^2} \left(\sqrt{x^2 + y^2} - R_{crt} \right)^2 \quad (5)$$

$$z_{crt}(x, y) = \frac{(x^2 + y^2)(D_{crt} - h_{crt})}{(R_{crt} - r_{crt})^2} - (D_{crt} - h_{crt}) \quad (6)$$

where, x and y are the x-direction and the y direction value of the Lunar

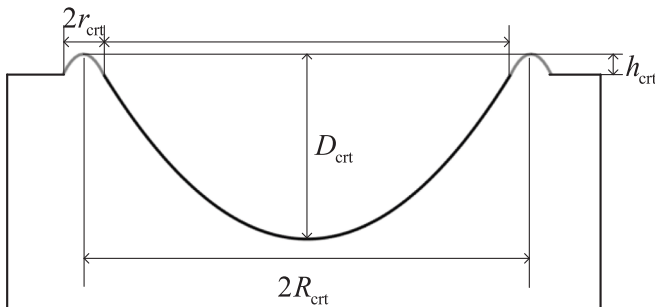


Fig. 2. Simplified Lunar crater model with the size parameters.

crater model; z_{crt} is the elevation value of the Lunar crater model; h_{crt} is the height of the crater rim; r_{crt} is the radius of the crater rim; D_{crt} is the crater depth; R_{crt} is the radius of the crater.

The shape of single crater is decided by the radius R_{crt} and the depth D_{crt} , and an indicator of **depth to diameter ratio (DDR)** can be used to describe the crater shape

$$DDR = \frac{D_{crt}}{2R_{crt}} \quad (7)$$

The craters with low DDR are flatter and shallower than those of higher DDR. Fig. 3 shows the craters with DDR of 0.05, 0.10, 0.15, 0.20 and 0.25 in the same radius.

Terrain complexity is defined as an index to evaluate the degree of fluctuation of the surface, which is used to describe the overall characteristics of terrain changes. The terrain complexity is composed of all the craters. Define the volume of all the craters in the terrain as V_{crt}

$$V_{crt} = \sum_{i=1}^{N_{crt}} \pi R_{crt}^2 D_{crt} \quad (8)$$

The volume of the craters V_{crt} is positive correlation with the number N_{crt} , the radius R_{crt} and the depth D_{crt} of the craters. Since, V_{crt} can roughly represents the terrain complexity.

The Lunar surface is littered with rocks ranging from centimeters to hundred meters in diameter and in different shapes. The rock model is represented by the curved surface obtained by the parabola rotating around the z-axis, which is calculated by the following formula

$$z_{rck}(x, y) = h_{rck} - \frac{4h_{rck}(x^2 + y^2)}{d_{rck}^2} \quad (9)$$

where, x is the x-direction value of the rock model; y is the y-direction value and z_{rck} is the z-direction value; h_{rck} is the height of the rock and d_{rck} is the width of the rock.

Depending on the number, size of craters on the Lunar surface, the terrain contains various topographic construct information. Though he DDR and volume describe the terrain complexity, they are sometimes hard to specifically measured through the 3D terrain data. An evaluation indicator, i.e **terrain frequency (TF)**, is proposed to describe the 3D terrain fluctuation imitating the image frequency. When modeling the Lunar surface environment, the projected height of the z-axis is converted into the intensity value of image pixels. The two-dimensional discrete Fourier transform is performed as

$$F(u_h, v_h) = \sum_{x=0}^{N_x-1} \sum_{y=0}^{N_y-1} f(x, y) e^{-j2\pi \left(\frac{u_h x}{N_x} + \frac{v_h y}{N_y} \right)} \quad (10)$$

$$= |F(u_h, v_h)| e^{-j\Phi(u_h, v_h)}$$

where, u_h, v_h are respectively the frequency of x direction and y direction (no unit), N_x and N_y are respectively the number of points contained in the x direction and y direction in a terrain map, $|F(u_h, v_h)|$ and $\Phi(u_h, v_h)$ are respectively the amplitude and phase at the point (u_h, v_h) .

Then the terrain frequency f_{ter} (unit: m^{-1}) can be defined as

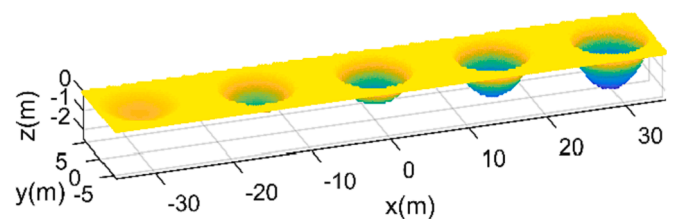


Fig. 3. Five craters with increased depth to diameter ratio (DDR) of 0.05, 0.10, 0.15, 0.20 and 0.25 in the same radius.

$$f_{\text{ter}} = \frac{\sum_{u_h} \sum_{v_h} \left(|F(u_h, v_h)| \cdot \sqrt{u_h^2 + v_h^2} \right)}{\sum_{u_h} \sum_{v_h} |F(u_h, v_h)| \cdot \sqrt{W_x^2 + W_y^2}} \quad (11)$$

where, W_x and W_y are the widths of x and y directions of the map respectively (unit: m).

2.3. LiDAR point cloud of Lunar terrain

After establishing the LiDAR perception model and the Lunar terrain model, a TIN (Triangulated Irregular Network) method is proposed to obtain point cloud data quickly and accurately. The essence of the proposed LiDAR perception model is to obtain the intersection of LiDAR lasers with environmental surface. However, current Lunar terrain information available are mostly points data, which is awkward for the calculation of intersection points. The discrete points data is transformed into continuous triangle surfaces information by the building of TIN-based DTM for the Lunar terrain, and offering the reliable and valid environmental information for output of LiDAR point cloud.

It is time consuming to calculate the intersection of each laser beam with each triangular surface. To shorten the computing time, this paper applies geometric constraints and multi-thread acceleration in the point cloud solution process. Regarding the LiDAR, the distribution of the laser beam is determined, and the point cloud solution can be accelerated by the distribution of the laser and the characteristics of TIN model, that is geometrical constraints, including horizontal constraints and vertical constraints. The horizontal constraints of all triangles are calculated by multi-thread, and the results of horizontal constraints are sorted to obtain the triangles set corresponding to each laser beam. Then, multi-thread acceleration is applied to select the triangles corresponding to each laser beam that satisfies the vertical constraints, and the intersection point is calculated by ray and triangle intersecting detection. Finally, the intersection points converge to form the Lunar point cloud.

2.4. Crater identification

The LiDAR point cloud obtained from TIN-based DTM can be used to perceive the Lunar environment to build an occupancy grid map and provide navigation information for rovers. Compared with the indoor environment, the rugged environment and complex terrain of moon make it difficult to accurately extract traversable areas. The main difficulty in perceiving the Lunar environment based on LiDAR point cloud is identifying negative obstacles (obstacles whose height is below the ground level), such as Lunar crater. Due to the scanning scheme of LiDAR, negative obstacles are difficult even almost impossible to be detected completely [22]. As shown in Fig. 4, the side of negative obstacle close to the LiDAR is in the detection blind area of LiDAR. LiDAR can only detect the partial area (the red line between p_1 and p_2) of

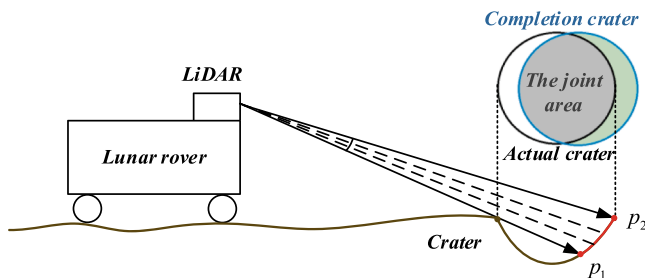


Fig. 4. Partial area (red line) of crater scanned by LiDAR. Because of the errors of the LiDAR and detection methods, the actual crater (black line) and the projection (green line) of the detected crater circle are not coincided. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the negative obstacle, so it is necessary to identify and complete the negative obstacles area to ensure the integrity of the traversable area. Hence, detection and completion of craters are the key steps to obtain the accurate occupancy grid map.

2.4.1. Detection of craters based on DBSCAN

Through the ground separation process, points which height is below the ground level is judged as negative obstacles. Since there are multiple negative obstacles in the environment, it is necessary to cluster them and obtain the point cloud data of each one. In general, clustering algorithms are divided into hierarchical and density-based algorithms. The former usually needs to know how many categories it needs to be divided into, while the latter does not. Since the number of negative obstacles is unknown, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) method is adopted to discover clusters of arbitrary shape (points of a single negative obstacle) based on a density-based notion of clusters [26]. Here, the clusters of arbitrary shape offer the edge points foundation for crater completion into circular shape.

It is important to evaluate the performance of the detection of the negative obstacles, which prevent the rover from entering the Lunar crater. The **detection probability (DP)** is proposed to evaluate the ratio of the detected craters to the all craters in the terrain.

$$DP = N_{\text{dte}} / N_{\text{all}} \quad (12)$$

where, N_{dte} is the number of the detected craters and N_{all} is the number of the all craters in the scanning range.

2.4.2. Completion for craters based on RANSAC

The circle completion is processed on each single cluster of the negative obstacle points after projecting the negative obstacle points onto the horizontal plane. The Random Sample Consensus (RANSAC) algorithm is used to fit the negative obstacle circle, which can iteratively estimate the parameters of a mathematical model from a set of observational data containing noise.

After the Lunar crater is detected and completed, the scene as shown in Fig. 4 will appear. The black circle represents the actual Lunar crater, and the blue represents the completed Lunar crater. After the completion process, the two circles will not completely coincide, resulting in the only partial completion of the Lunar crater. The rest part of crater would be recognized as traversable area, leading to danger. Therefore, the **completion rate (CR)** of Lunar crater is proposed to evaluate the effect of Lunar crater completion algorithm.

$$CR = S_{\text{com}} / S_{\text{crt}} \quad (13)$$

where, S_{com} is the joint area of the completion crater and the actual crater and S_{crt} is the area of the actual crater.

The completed craters are taken account into the occupancy grid map for autonomous navigation. Compared to the point cloud by LiDAR, the occupancy grid map after completion of Lunar craters provides the safer area for Lunar rover more entirely, and protects the rover from falling into the obstacles.

Section 2.1 introduces the LiDAR model, using density and precision to describe the point cloud quality. Section 2.2 introduces the terrain simulation method and puts forward the terrain frequency to describe the different complexity caused by the number, width and depth of the crater. Section 2.3 introduces the point cloud simulation method. Section 2.4 introduces the method of crater identification based on LiDAR point cloud, which is divided into two steps: crater detection is described by detection probability, and crater completion is described by completion rate. Both terrain fluctuation and point cloud quality cause changes in environment perception performance. To be specific, terrain frequency, point cloud density and point cloud accuracy influence detection probability and completion rate. The influence mechanism is studied in the simulation chapter.

3. Simulation and results

At present, the core problem is that the influence mechanism of terrain and point cloud on environmental perception is not clear. To clarify this problem, the following simulation experiment is designed. Firstly, the rationality and validity of topographic frequency should be verified, and the mapping relationship between terrain frequency and the number, width and depth of meteorites should be proved. Then, the effects of topographic frequency, point cloud quality (density, ranging error, horizontal angle error and vertical angle error) on the DP and CR of crater identification are studied. Thus, the performance design of LiDAR system under different terrain scenarios in the application of lunar rover can be guided.

3.1. Environmental simulation of Lunar spatial data

Firstly, the simulation terrain is formed based on the Lunar terrain model in the study results of NASA [27], including the shapes with depth and rim height data for various age groups illustrated in the Table 1, and the range of crater cumulative frequency distributions for the Lunar terrains shown in Fig. 5. The shapes and sizes of the craters in the simulation terrain follow these statistical distributions.

The Lunar surface environment of $100\text{ m} \times 100\text{ m}$ is established. Then, based on the established mathematical model of craters and rocks with the statistical analysis of Lunar topographical data, corresponding number and size of craters and rocks are generated as shown in Fig. 6. The crater type is selected obey the Table 1 and selected from the all three types. Fig. 6 shows an example of terrain simulation that uses a number of different terrains in Section 3.

In order to verify that the terrain frequency is able to comprehensively represents the number, width and depth of the craters and then evaluates the terrain complexity, we firstly analyze the terrain frequency under the varied physical quantities of the craters. In a flat terrain of the same size, different numbers of Lunar craters of the varied size are added to form unstructured terrain with varying complexity. The terrain frequency is influenced together by the number and the radius of the craters and it is increased with the rising of the number and radius of the craters. According to the number and radius, the volume of craters on the Lunar surface is calculated, and the relationship between the terrain frequency and the volume is analyzed. As shown in Fig. 7, the terrain frequency is nearly linear correlation with the volume. In addition, the same volume with different number of the craters, the more craters the larger terrain frequency, indicating that the number contributes more than the size to the terrain complexity. In conclusion, the terrain frequency is able to comprehensively describe the terrain fluctuation when the craters volume is difficult to measure.

3.2. Perception analysis of imperfect crater

In real terrain, the crater shape is imperfect, maybe there are some small rocks inside or around the edges of a crater, as shown in Fig. 8.

By changing the size ratio and height ratio of edge-rocks and crater, the crater recognition results are shown in Fig. 9. When the center of the rock is located at the edge of the crater, the ratio of the size of the rock to the radius of the crater is less than 84.95 %, the completion rate of the crater reaches more than 90.07 %.

In summary, the crater detection and completion method proposed in this paper performs well even when there are small rock in the crater or

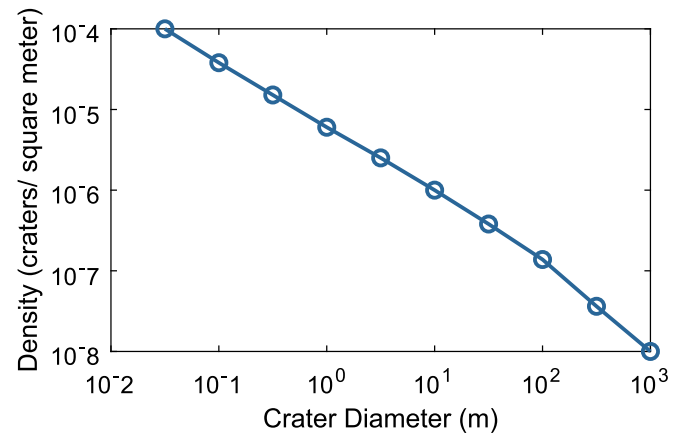


Fig. 5. Cumulative crater distribution with number of craters larger than diameter for the Lunar terrains [18].

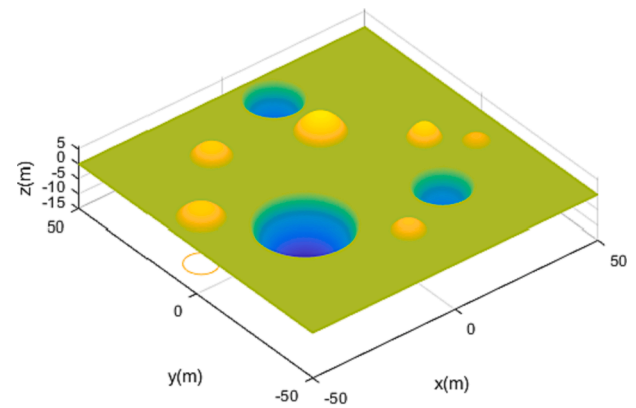


Fig. 6. Topographic model of the Lunar terrain containing the craters and the rocks.

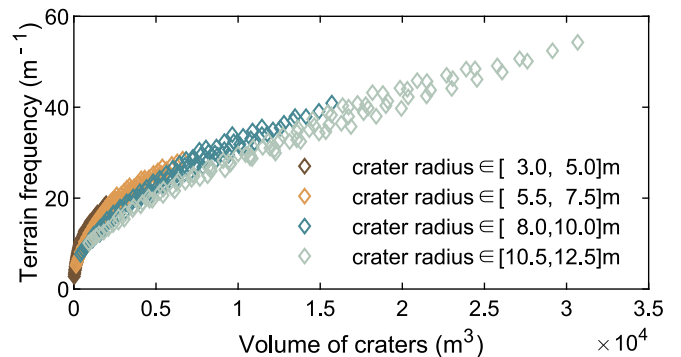


Fig. 7. The terrain frequency is nearly linear correlated to the volume of craters, and the same volume with smaller size makes the higher terrain frequency.

on the edge, reflecting the robustness and universality of the algorithm.

3.3. Perception analysis of crater detection

In actual environment perception, the rocks and the craters are both obstacles hindering rover travelling. As positive obstacles, the rocks are easy to separate from environment for obstacles identification. However, craters as negative obstacles need to be completed for lack of partial points close to sensor. Thus, this work focused on detection and completion of the craters.

Table 1

Idealized Crater Shapes in relation to morphology [27].

Crater Type	Depth to Diam Ratio	Rim Height to Diam Ratio
Fresh	0.23 to 0.25	0.022 to 0.06
Young	0.17 to 0.19	0.016 to 0.045
Mature	0.11 to 0.13	0.008 to 0.03

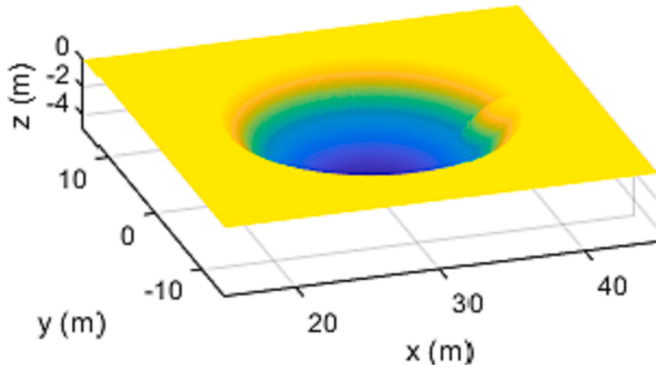


Fig. 8. The terrain map of imperfect crater with rocks.

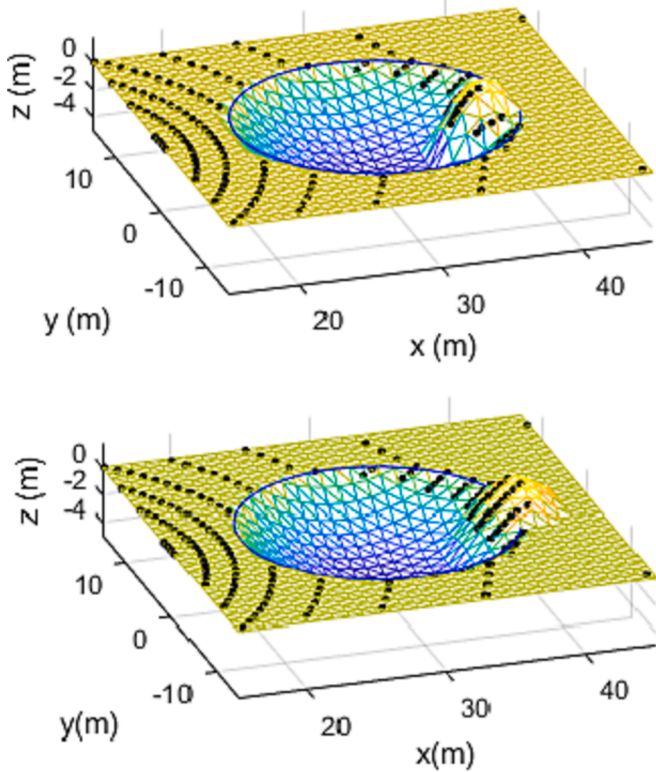


Fig. 9. The LiDAR point cloud of imperfect crater with rocks.

In this section, the aim is to change the point cloud density and accuracy and simulate its detection probability of crater detection by varying the system parameters of the LiDAR perception model. First of all, the default setting of the rover system based on LiDAR is, the number of pixels of in one frame with a FOV of $16^\circ \times 60^\circ$ determined as 32×128 , with the angular intervals of 0.5° , the installation height of LiDAR is 1.5 m. In this study, a crater with the size of 2 m–20 m could be detected from a distance of 50 m.

The influence of point cloud density on crater detection under certain point cloud accuracy is simulated. The number of horizontal and vertical pixels, angular resolution and crater distance from LiDAR origin were varied to obtain different point cloud density conditions ranging from 0.898 pts/m^2 to 8.989 pts/m^2 . The pointing error and ranging error were 0. The results of DP are shown in Fig. 10. In order to increase 17.7 % DP, it is necessary to increase the number of horizontal or vertical pixels by twice, and the point cloud density increases from 0.898 pts/m^2 to 1.796 pts/m^2 . However, when the point cloud density is greater than 1.796 pts/m^2 , which was regarded as lower limit point density (LLPD),

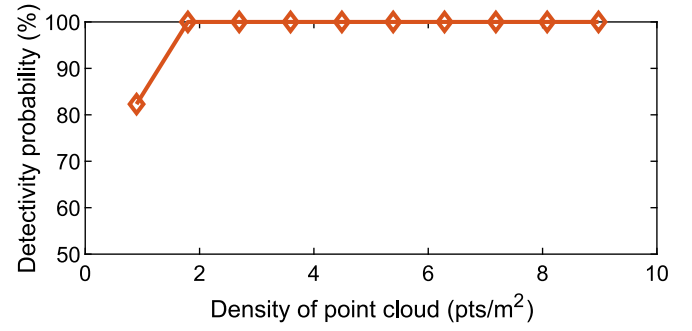


Fig. 10. Detection probability increases with the rising of the density of point cloud until the density became greater than lower limit point density (LLPD).

the point cloud density has no obvious enhancement effect on DP.

But in practice, LiDAR system contains pointing errors and ranging error. Due to the ranging error ΔR_{ij} and the pointing errors $\Delta \theta_i, \Delta \varphi_j$, point cloud generated from the LiDAR perception model differs from the actual terrain, which influence the accuracy of the point cloud. The ranging error and pointing errors may leads to the crater point to be mistaken as ground point, which cannot be clustered as a crater, resulting in the decrease of DP.

The basis of crater detection is that the elevation of negative obstacle is smaller than the Lunar surface, and the accuracy of ranging and vertical angle affect the elevation accuracy of negative obstacle point cloud, so the negative obstacle detection rate is greatly affected, as shown in Fig. 11. However, the horizontal angle error has little effect on the negative obstacle DP. When the vertical error is greater than 0.3° and the ranging error is greater than 0.07 m, the DP will be less than 80 %. If the DP of negative obstacle is less than 80 %, it can be considered as a relatively dangerous situation. Therefore, an error optimization range can be determined according to this index.

After analyzing the influence of single error on crater detection, the coupling effect of error on crater detection is further discussed. Since the horizontal angle error has little effect on the detection probability, the horizontal angle error is ignored in this simulation. The ranging error and vertical angle error are changed to obtain the DP under the combined change of both, as shown in Fig. 12. Seen from Fig. 12, in these two error ranges within the ranging error from 0 to 0.15 m and the vertical angle error from 0 to 1.5° , the DP is more affected by the vertical angle error. When the vertical angle error is greater than 0.3° , the DP drops sharply from 80 %. To meet the requirements of Lunar rover safety, the maximum combination of ranging error and vertical angle error can be determined by the critical indicators of DP.

3.4. Perception analysis of crater completion

Similar to the previous section, the simulation of the effect from point cloud density and accuracy on completion rate is carried out. The DP is from a macro point of view to the total craters, while the CR is considered from a microscopic perspective for the single crater. The points obtained by LiDAR almost do not coincide with the actual edge of the crater, it is almost impossible to completely fit the actual coverage of the crater. Therefore, unlike the DP, due to the discrete points from LiDAR, the CR would hardly reach 100 %.

At the first, the CR is affected by the geometric properties of the crater, and mainly increases with the grow depth to diameter ratio (DDR), shown as Fig. 13. That's because as the DDR grows, more points fall on the crater edge to provide more shape information to complete the crater, as shown in Fig. 14. Therefore, according to the statistic the DDR of the craters in the Lunar surface is between 0.11 and 0.25, so the range of the CR is from 70 % to 90 %, which offers instruction for crater expansion.

As the size properties of the crater modifies of terrain frequency, the

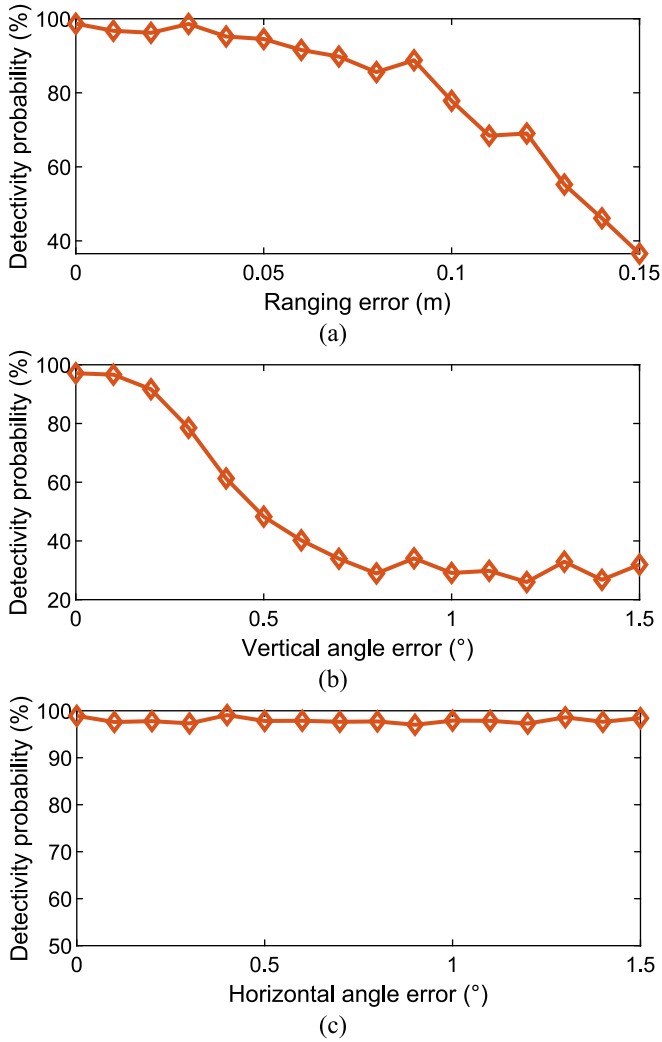


Fig. 11. Detection probability under different error level of ranging distance, vertical angle and horizontal angle. (a) DP decreases with the ranging error; (b) DP decreases with the vertical angle error; (c) DP is little affected by the horizontal angle error.

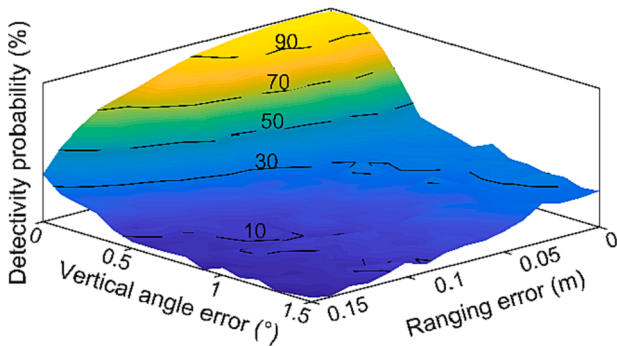


Fig. 12. Detection probability is influenced by ranging error and vertical angle error.

above experimental results can be represented as the influence of terrain frequency on CR, as shown in Fig. 15. The change of terrain frequency caused by the size of crater effect crater identification micro representation - CR. The terrain frequency and CR do not correspond one to one, but the CR shows a general trend of increasing with the rise of terrain frequency. The environment with high terrain frequency provides more

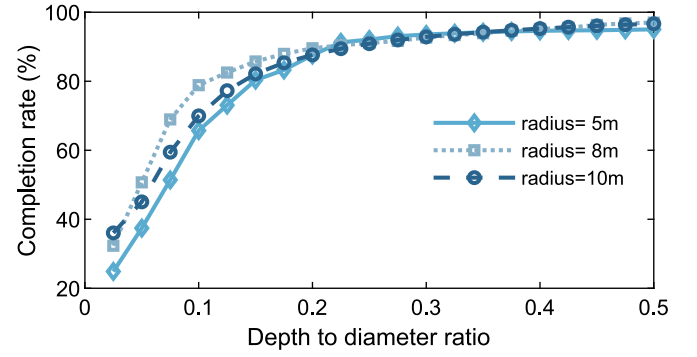


Fig. 13. Completion rate increased with the rising of the depth to diameter ratio (DDR) and independent of the radius.

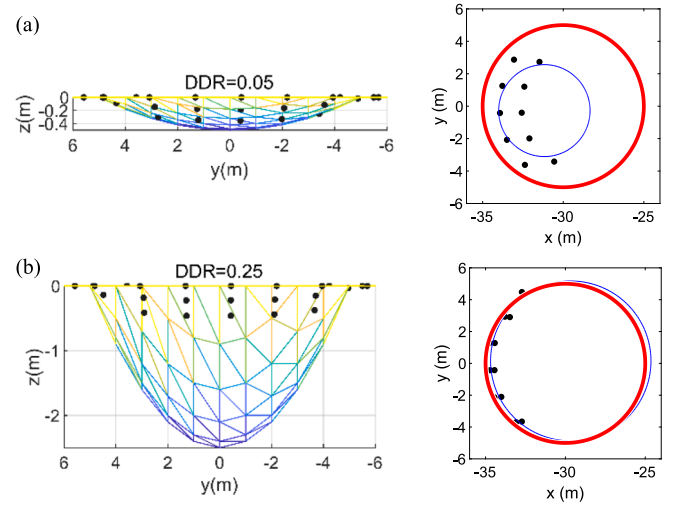


Fig. 14. Comparison of the craters between different depth to diameter ratio (DDR) (0.05 in (a) and 0.25 in (b)), completion rate of 37.4% in (a) and 92.1% in (b). Where, the black points mark the point cloud obtained by the LiDAR, the red circle represents the actual crater, and the blue circle represents the completion crater. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

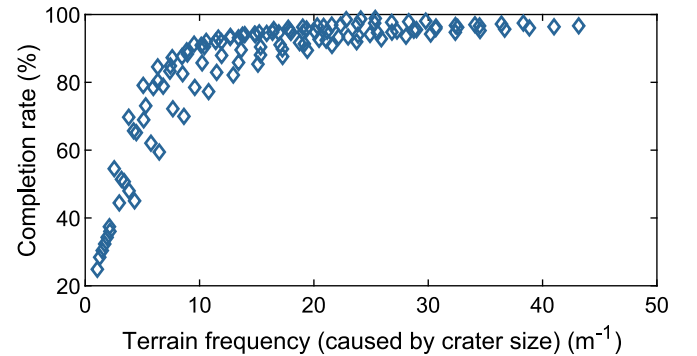


Fig. 15. Completion rate under different terrain frequency varied by crater size.

sufficient information for the crater identification of LiDAR point cloud, while the environment with low terrain frequency has a low danger degree despite of low CR. Without the LiDAR systematic errors, when the terrain frequency is greater than 20.10 m^{-1} , the CR is more than 90 %. In addition, the DP of crater is basically independent of the terrain frequency. It can be concluded that this crater detection and completion

method can realize safe crater identification under different terrain conditions.

When the terrain frequency is larger than 20 m^{-1} , the influence mechanism of point cloud accuracy on CR is studied. As for the influence of point cloud accuracy on CR, as long as the negative obstacle point cloud can be detected and the clustering is realized, the pointing error no matter horizontal angle or vertical angle, has almost no influence on the CR of the negative obstacle. This is because the pointing error does not significantly change the shape of the negative obstacle edge. Whereas, the increase of ranging error destroys the edge contour information of negative obstacle, resulting in the reduction of CR, as shown in Fig. 16.

Compared with the DP, the low CR can be compensated by the expansion operation of negative obstacles. Therefore, DP is more important for the identification of negative obstacles. Therefore, the density and accuracy requirements of point cloud should be determined according to the detection probability within the safety limit, so as to optimize the LiDAR system parameters and error range.

4. Discussion

Through the proposed crater detection and completion method, the craters can be reasonably identified under different terrain conditions, the terrain complexity and the quality characteristics of LiDAR point cloud are studied to determine the projection relationship between influencing factors and crater identification performance. The simulation results emphasized that, ranging error and vertical angle error are the dominant influencing parameters that change point cloud quality and affect environment perception performance. Through the evaluation of the crater identification experiments, three points were noted:

1. **Terrain Frequency (TF):** An indicator on frequency domain to comprehensively contains the terrain complexity. TF is linear correlation to the volume of the craters which is the product of the number, the radius and the depth of the craters. It is important to use TF describe the terrain as the obstacles volume is difficult to measure. TF mainly affects the performance of crater completion. When TF is greater than 20 m^{-1} without LiDAR systematic errors, the completion rate reaches more than 90 %.
2. **Lower limit point density (LLPD):** LLPD is an important density value for the probability of crater detection. Decreasing points density results in performance degradation of crater detection. For a crater 15 m away, LLPD is 1.7957 pts/m^2 , which means that when the point density is greater than 1.7957 pts/m^2 , the increase of point density has no significant effect on the crater detection rate. At this time, the number of pixels of the LiDAR with a FOV of $17^\circ \times 120^\circ$ can be determined as 32×64 . LLPD can be used as a balance reference between performance and computational cost of environment perception.
3. **Ranging error and vertical angle error:** Ranging error and vertical angle error are the major influence factors of the LiDAR point cloud accuracy, respectively dominating in completion rate and detection probability. To meet the requirements of Lunar rover safety, the optimal combination of ranging error and vertical angle error can be determined by the critical indicators of crater identification performance. As the range error is less than 0.07 m and the vertical angle error less than 0.3° , the detection probability is higher than 80.58 %, and the completion rate is higher than 84.18 %.

These experimental results provide researchers a useful guide to find the dominant influencing factors in crater identification for Lunar rover traversing in the future. The density and accuracy requirements of point cloud should be determined according to the detection probability and completion rate within the safety limit, so as to optimize the LiDAR systematic error range.

It should be noted that the terrain data used in this paper is the simulated data of crater models. There are two reasons for this. Firstly,

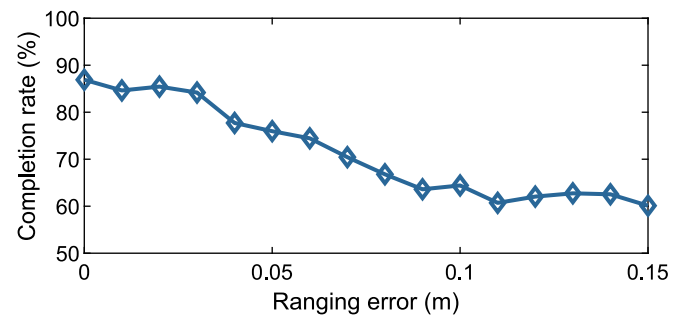


Fig. 16. Completion rate decreases with the rising of the ranging error.

to study the influence mechanism of terrain complexity and LiDAR system parameters on environment perception, the simulated data can provide the change of crater size, number, point cloud density and other parameters. Secondly, the realistic crater data are DTM-based crater data collected from orbit that is oversize for rover route planning, thus the simulated data are needed to provide the “rover obstacles avoidance” scale crater data for further analysis.

5. Conclusion

To identify a pattern of partial crater in the lunar terrain, a three-dimensional environment perception method based on LiDAR installed on the Lunar rover was developed. The relations between the identification of craters and terrain frequency, ranging error and vertical angle error are investigated. The terrain frequency is nearly linear correlated to the volume of craters. The crater detection and completion algorithm is put forward based on DBSCAN and RANSAC, to improve the safety performance of driving in the complex lunar terrain, which are mainly evaluated by completion rate and detection probability of negative obstacle area in occupancy grid map, respectively.

The experimental results of simulations show that the ranging error and the vertical angle error are main factors influencing the detection probability, and terrain frequency and ranging error are the major influential factors of the completion rate, that are two indicators at the macro level and the micro level to reveal the performance of crater identification. When terrain frequency is higher than 20 m^{-1} , the ranging error lower than 0.07 m and the vertical angle error lower than 0.3° , the acceptable detection probability and completion rate of 80.58 % and 84.18 % can be obtained. In order to improve the authenticity of simulation, the rocks-and-crater influence on rover environment perception will be considered in the lunar surface topography in the future work. In future work, other aspects of environment perception under different terrain conditions and point cloud quality, such as path planning performance, will be evaluated.

CRedit authorship contribution statement

Yier Zhou: Methodology, Validation, Formal analysis. **Xiaolu Li:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing. **Baocheng Hua:** Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research work was supported in part by Lab of Space Optoelectronic Measurement & Perception (LabSOMP202101); funded by Science and Technology on Space Intelligent Control Laboratory (2021-JCJQ-LB-010-05).

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